Timeline: Adding the time dimension to spreadsheets

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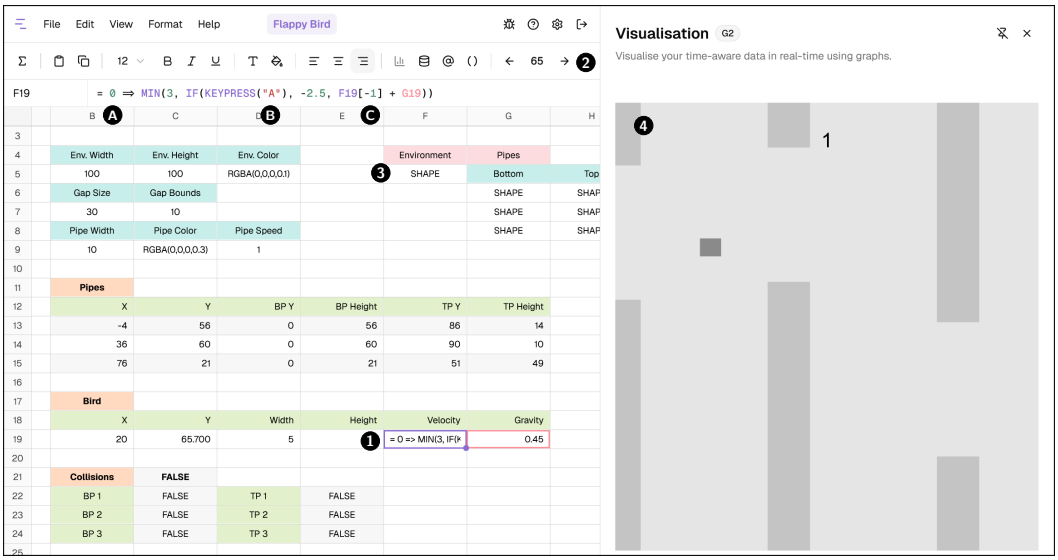

Fig. 1. A flappy bird game clone in Timeline. ❶ The selected formula computes the Y velocity in the current step. ❷ The system uses discrete time. ❸ Visualizations are built using a composable grammar of graphics and ❹ can be displayed inline or in a separate panel. The formula for the velocity Ⓐ specifies an initial value, Ⓑ reacts to user-interface events, and Ⓒ accesses a previous value to compute the current one.

Spreadsheets make it easy to express computations over two-dimensional data, but two dimensions are not enough to express rich computations with time such as simulations, complex systems models, analyses of historical data, or interactive systems. We present Timeline, a system that adds discrete time to spreadsheets.

The remarkable insight from our work is that a large number of advanced programming language concepts can be directly applied in the context of spreadsheets with time. Timeline draws from *dataflow languages* to express computations over time, *coeffect systems* to store only the necessary previous values, *functional reactive programming* to support interactivity, *grammars of graphics* to support composable visualizations, and *typed holes* for inserting cell references in the formula editor.

In this paper, we provide an overview of the Timeline design and discuss how it adapts the aforementioned programming language innovations for the context of spreadsheets. We then formalize the evaluation of spreadsheets with discrete time through a *core calculus*, describe a coeffect-based inference for computing the required number of past values and prove that the optimization is sound. More broadly, this paper shows that established programming language ideas can often be productively used outside of their original domain.

CCS Concepts: • **Software and its engineering → Very high level languages**.

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1 Introduction

Spreadsheets are sometimes referred to as "the most widely used programming systems in the world" [2]. Their uses range from personal productivity tools and decision making support systems to scientific analytical instruments and financial reporting systems [11]. However, spreadsheets are first-order, limited to tabular data structures, and lack abstraction mechanisms.

Many extensions to the core spreadsheet model have been proposed, generalising the tabular structure to more general forms [3, 7], adding support for objects [19], or incorporating support for streaming data to use spreadsheets as the back-end for web development [8, 9]. Abstraction mechanism developed for spreadsheets include sheet-defined functions [16, 24] and gridlets [15], while programming by example [12] simplifies data wrangling within spreadsheets.

As the above extensions show, extending spreadsheets is a difficult balancing act. How can we make spreadsheets more expressive, while preserving their liveness, directness and usability [14]? The aim of the Timeline project is to extend spreadsheets with a built-in notion of discrete time, making it possible to use spreadsheets in a range of domains that are difficult to tackle with only two dimensions [17] such as physics simulations, agent-based modelling [1], financial data analysis, or simple interactive applications. As illustrated in Figure 1, Timeline adds an implicit discrete time to the spreadsheet programming model and extends the formula language to allow references to previous values and user interface events, but otherwise preserves the standard spreadsheet model.

Adapting programming langauge research. Although Timeline is interesting as an exploration of the expressive power of spreadsheets with discrete time, this paper considers it primarily from the technical perspective of programming language design and implementation. We show that a large number of concepts and techniques developed in the context of programming language research are directly applicable to spreadsheets.

The key addition to Timeline is a built-in notion of discrete time. Formulas in cells can refer to other cells (e.g. `G19`), but also values computed in earlier steps (e.g. `F19[-1]`). The language design and runtime implementation of Timeline draws inspiration from research on *synchronous dataflow languages* [4, 13]. To ensure that simulations and interactive applications execute using bounded memory whenever possible, we augment the runtime with a program analysis based on *coeffects* [22, 23]. The support for discrete times also makes it possible to use the spreadsheet paradigm for building interactive applications. Here, we draw on the *functional reactive programming* paradigm [10, 25]. Although not strictly related to time, Timeline also adopts a *composable data visualization* model [18, 21, 26]. Finally, the Timeline also uses a notion of *typed holes* [20] to determine when the user may want to insert a cell reference during formula editing.

Contributions. This paper presents the design and implementation of a spreadsheet system with discrete time and captures its essence through a core calculus.

- We provide an overview of the Timeline system design through a simple physics simulation example (§2) and then examine the expressive power of the programming model through a number of case studies covering agent-based modelling, simulations and finance (§5).
- We discuss how how the design and implementation of Timeline adapts a range of advanced programming language concepts and techniques for the context of spreadsheet systems (§3), including dataflow languages, coeffects, functional reactive programming, visualization grammars and typed holes.
- We present a core calculus of the Timeline execution model, which captures the way previous values are stored during the spreadsheet evaluation (§4). We formalize a space optimization based on coeffects implemented in Timeline and prove its soundness (§4.1).

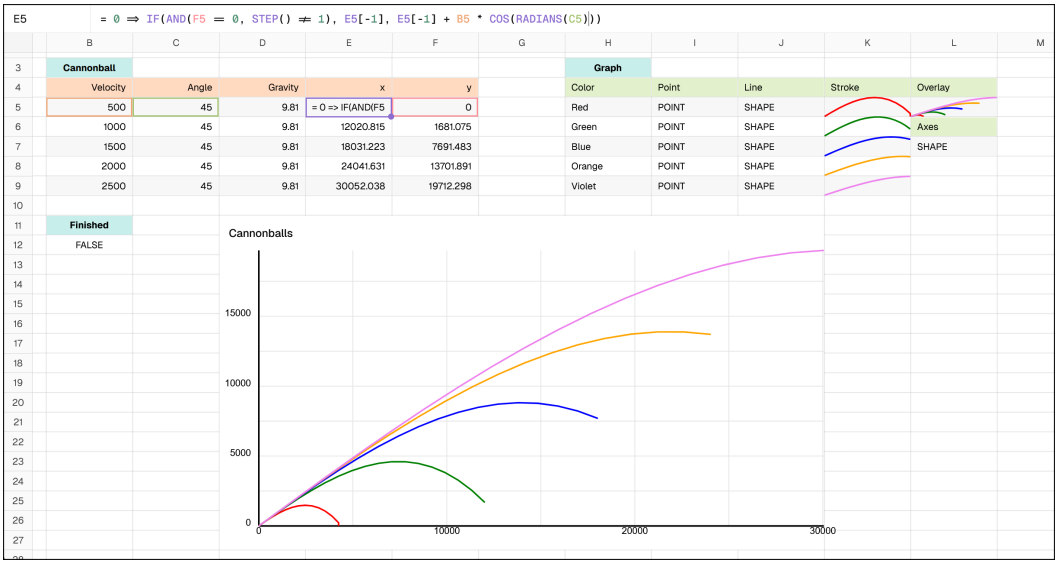

Fig. 2. Cannon ball.

2 Overview

We overview Timeline using a simple cannonball firing simulation (Figure 2). In each step, the simulation computes the cannonball position using the position in the previous step and the current time step. We ignore air resistance, but this could be easily added to a stepwise simulation model. The simulation stops when the cannonballs hit the ground.

Cannonball movement. bar

```
E5 = 0 => IF(AND(F5 == 0, STEP() != 1), E5[-1], E5[-1] + B5 * COS(RADIANS(C5)))
F5 = 0 => MAX(0, F5[-1] + B5 * SIN(RADIANS(C5)) - 1/2 * D5 * POWER(STEP(), 2))
```

3 Language features

3.1 Synchronous languages

3.2 Coeffects

3.3 Functional reactive programming

3.4 Composable data vis

3.5 Typed holes

4 Calculus

4.1 Coeffect soundness

5 Case studies

5.1 Formative interviews

6 Related work

sheet defined functions [16] [24] and their semantics [6]

gridlets [15] and their semantics [27] [5]

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